

Experimental Tutorial #1

1. Safety notes

- No eating or drinking inside the laboratory.
- Avoid touching heated or electrically active components.
- Ensure hands are dry and avoid placing metal objects near powered circuits.
- Do not modify wiring while the supply is ON.
- Report damaged cables or unusual odors immediately.
- Maintain a clean, organized workspace.

2. Introduction

This laboratory session investigates the operation of a series linear voltage regulator using a two-transistor closed-loop architecture, representative of a classical analog feedback system. Students will perform direct measurements on the completed circuit and analyze how negative feedback improves regulation performance.

In the second part of the tutorial, the Zener diode is characterized experimentally to understand its non-ideal behavior and dynamic resistance.

3. Learning objective

By the end of this tutorial, you will be able to:

Regulator Behavior

- Understand how negative feedback stabilizes output voltage.
- Measure line regulation S_{V_i} and load regulation (output resistance) S_{I_L} of a closed-loop linear regulator.
- Observe how the feedback loop reacts to disturbances.

Reference Diode Characterization

- Identify the Zener breakdown operating region.
- Measure the Zener voltage and current under different bias conditions.
- Extract the dynamic resistance r_Z and relate it to regulator performance.

4. Required equipment

- Digital multimeter (voltage, resistance)
- DC power supply (12–15 V range)
- Pre-assembled laboratory board with the two-transistor feedback regulator
- Zener diode ZPD5V1 and series resistor (for Part 2)
- Hook-up wires

1. Closed-Loop Two-Transistor Voltage Regulator

5. Circuit overview

The circuit under test is the series-parallel negative-feedback regulator made of:

- Q1 (BD139): series pass transistor
- Q2 (2N2222): error amplifier comparing feedback voltage vs. Zener reference
- Zener diode providing the reference voltage
- Feedback divider (R_3 – R_4) sampling the output
- Input capacitors improving stability

In this circuit, the output voltage V_o is sampled by a voltage divider formed by R_3 and R_4 , establishing a feedback factor $\beta = \frac{R_4}{R_3 + R_4}$. The output from the divider serves as the feedback voltage which is "subtracted" from the control voltage (i.e., the V_Z across the Zener diode) through transistor Q_2 . The collector voltage of Q_2 thus provides the input voltage V_i to our unity-gain follower amplifier. Capacitor C_1 is included to narrow the amplifier's bandwidth, enhancing stability.

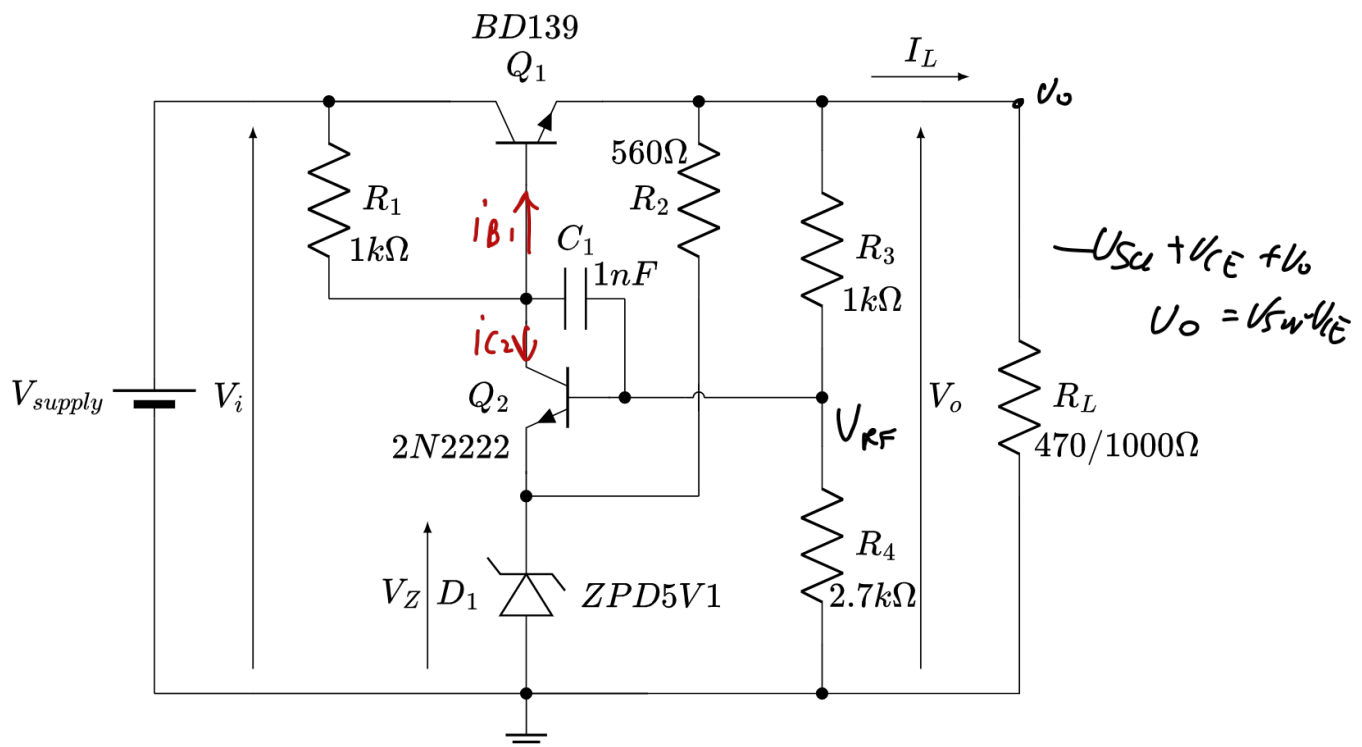
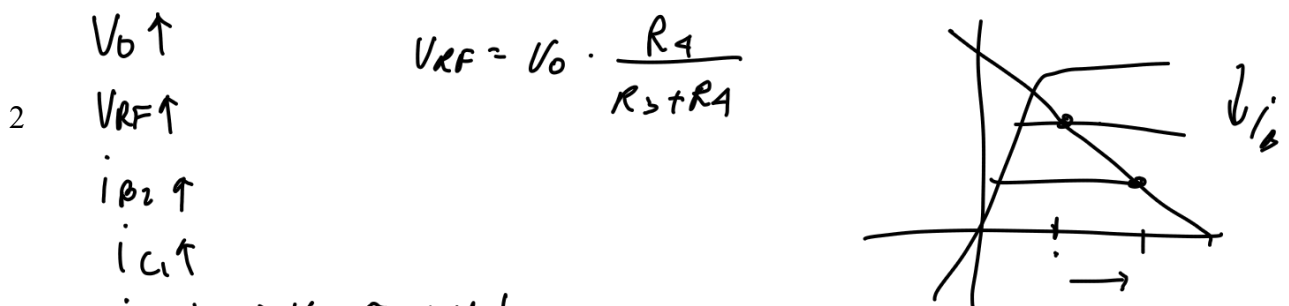


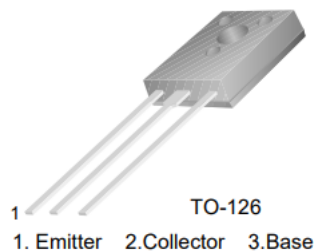
Fig. 1 – Voltage regulator circuit with negative feedback.



$$I_B \downarrow \rightarrow V_{CE} \uparrow \rightarrow V_o \downarrow$$

2N2222; 2N2222A

BD135 / 137 / 139 NPN Epitaxial Silicon Transistor



PINNING

PIN	DESCRIPTION
1	emitter
2	base
3	collector, connected to case

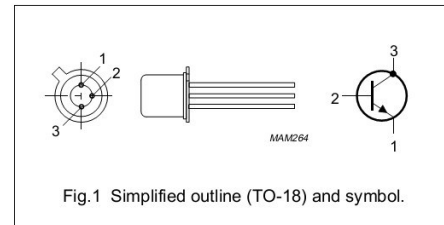


Fig. 2 – Pin configuration of the BD139 (left) and of 2N2222 (right) transistors.

6. Measurements

Objective:

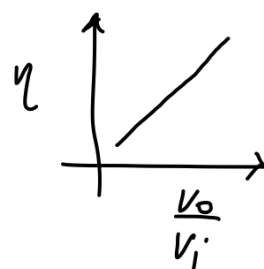
Quantify how the output voltage changes when the input voltage varies.

Procedure:

1. Set a fixed load resistor and measure its value through the multimeter.
2. Apply 12 V at the input and measure: input voltage output voltage, Zener voltage (record all the displayed digits).
3. Increase the input to 15 V and repeat the measurements.
4. Change the resistor value and repeat from point 2.

Calculation:

1. Evaluate the input voltage sensitivity as: $S_{V_i} = \frac{\Delta V_o}{\Delta V_i}$, expect a small value for the feedback regulator.
2. Compute the load current through the Ohm's law.
3. Evaluate the regulator output resistance (load current sensitivity) $S_{I_L} = \frac{\Delta V_o}{\Delta I_L}$
4. Compute the efficiency $\eta = \frac{V_o}{V_i}$



V_i	R_L	V_z	V_o	I_L	η
11,91V	470	5,09	7,88 V	16,77 mA	0,661
15,16V	470	5,11	7,97 V	16,95 mA	0,526
11,91V	1 kΩ	5,10	7,88 V	7,88 mA	0,661
15,16V	1 kΩ	5,11	7,95 V	7,95 mA	0,524

$$S_{V_i} = \frac{\Delta V_o}{\Delta V_i} = \underline{0,0277} \text{ [V/V] with } R_L = \underline{470} \Omega$$

$$S_{V_i} = \frac{\Delta V_o}{\Delta V_i} = \underline{0,0215} \text{ [V/V] with } R_L = \underline{1000} \Omega$$

$$S_{I_L} = \frac{\Delta V_o}{\Delta I_L} = \underline{0} \text{ [Ω] with } V_i = \underline{11,91} \text{ V}$$

$$S_{I_L} = \frac{\Delta V_o}{\Delta I_L} = \underline{2,22} \text{ [Ω] with } V_i = \underline{15,16} \text{ V}$$

2. Zener Diode Characterization

7. Circuit overview

The circuit to build is shown in Figure 3. It is a simple Zener diode biasing network, consisting of the variable panel power supply, a resistor, and the diode. Figure 4 shows the main characteristics of the Zener diode.

The goal of this phase of the exercise is to verify the breakdown condition of the Zener diode and to measure the voltage across the Zener and the current flowing through it as the power supply voltage varies from +12V to +15V. This allows us to determine the Zener's dynamic resistance under breakdown conditions.

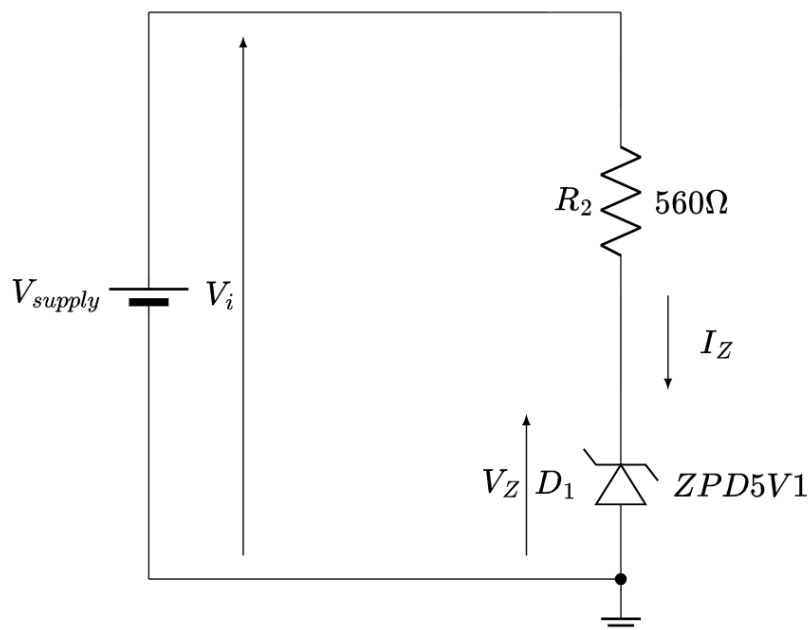


Fig. 3 – Circuit for the measurement of the Zener diode's dynamic resistance.

Power Dissipation at T _{amb} = 25°C		P _{tot}	500 ⁽¹⁾		mW
Junction Temperature		T _j	175		°C
Thermal Resistance Junction to Ambient Air		R _{thJA}	–	–	0.3 ⁽¹⁾ °C/W

Type	Zener Voltage ⁽¹⁾ at I _Z = 5 mA V _Z (V)	Dynamic Resistance		Temp. Coeff. of Zener Voltage at I _Z = 5 mA α _{vz} (10 ⁻⁴ /K)	Reverse Voltage at I _R = 100 nA V _R (V)	Admissible Zener current ⁽²⁾	
		at I _Z = 5 mA f = 1 kHz r _{zj} (Ω)	at I _Z = 1 mA f = 1 kHz r _{zj} (Ω)			at T _{amb} = 45°C I _Z (mA)	at T _{amb} = 25°C I _Z (mA)
ZPD5.1	4.8 ... 5.4	30 (< 60)	< 480	-3 ... +4	> 0.8	67	80

Fig. 4 – Zener diode's characteristics (ZPD5V1).

8. Measurements

Objective:

Quantify the Zener's dynamic resistance.



Procedure:

1. Measure the value of R_1 the multimeter.
2. Build the circuit as in Fig. 3.
3. Apply 12 V at the input and measure: the input voltage and the Zener voltage (record all the displayed digits).
4. Increase the input to 15 V and repeat the measurements.

Calculation:

1. Evaluate the voltage across R_1 and calculate the current through the Ohm's law.
2. Evaluate the Zener dynamic resistance as $r_Z = \frac{\Delta V_Z}{\Delta I_Z}$

V_i	R_1	V_Z	I_Z
11,91 V	565 Ω	5,12 V	$\frac{11,91 - 5,12}{565} = 0,012 \text{ A}$
15,48	569 Ω	5,15 V	$\frac{15,48 - 5,15}{569} = 0,01828 \text{ A}$

$$r_Z = \frac{\Delta V_Z}{\Delta I_Z} = \frac{5,15 - 5,12}{0,01828 - 0,012} [\Omega] = 4,77 \Omega$$

